

Applied Data Analytics Capstone Project

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Project Overview

The Applied Data Analytics Capstone Project serves as the culminating experience of the programme. It enables learners to systematically apply concepts covered across Excel analysis, Python-based exploratory analysis, SQL querying, Power BI dashboards, inferential statistics, and simple linear regression within a structured project environment.

The project mirrors a real-world analytical workflow where teams move step by step — from dataset selection and cleaning to business problem framing, structured analysis, statistical validation, regression modelling, and final presentation. Each phase is designed to build analytical clarity, technical discipline, and documentation rigour.

The workflow is deliberately organised across multiple sprints to ensure progressive development. Learners first understand and prepare the dataset, then explore and validate insights using analytical tools, and finally translate results into actionable business recommendations supported by clear storytelling and confident presentation.

In addition to technical correctness, the capstone emphasises structured thinking, hypothesis-driven analysis, dashboard narration, and viva defence. Learners are expected to demonstrate not only what they analysed, but also why their approach is methodologically sound and aligned with stakeholder objectives.

Key Tasks for Learners

As part of the capstone journey, each learner team will:

- Select a problem theme and finalise an appropriate dataset.
- Select a project topic and dataset from the approved list and understand evaluation criteria.
- Perform structured data cleaning, create documentation, and maintain version control using Git.
- Conduct Excel analysis, Python-based EDA, SQL querying, Power BI dashboarding, statistical validation, and simple linear regression analysis.
- Translate analytical outputs into business insights, recommendations, limitations, and future scope.

For each learner team, the sub-tasks include:

1. **Structured Data Preparation**
Explore dataset structure, handle missing values and outliers, create a data dictionary, and store cleaned data in SQL/CSV/Excel formats with version control.
2. **Analytical Exploration & Dashboard Creation**
Apply Excel functions, pivot tables, descriptive statistics, Python-based EDA, SQL aggregations, and Power BI dashboards to uncover patterns and define KPIs.
3. **Statistical Validation & Regression Analysis**
Perform probability calculations, conduct z-test and t-test, interpret statistical outputs, build a simple linear regression model, generate predictions, and evaluate results using appropriate metrics.
4. **Business Interpretation & Presentation**
Consolidate findings into structured business insights, draft recommendations,

identify limitations and future scope, compile final documentation, and prepare for showcase and viva sessions.

By the end of the project, learners will have:

- Executed a complete end-to-end analytical workflow across structured sprint phases.
- Developed disciplined documentation practices including preprocessing logs and Git repositories.
- Applied Excel, Python, SQL, Power BI, inferential statistics, and simple linear regression in an integrated manner.
- Translated statistical and regression outputs into stakeholder-aligned recommendations.
- Built confidence in presenting findings and defending methodology during viva sessions.

The capstone is not merely a technical assignment; it is a structured demonstration of analytical reasoning, tool proficiency, collaborative execution, and professional communication. It prepares learners to approach data-driven problems with clarity, methodological rigour, and accountability.

Project Objectives

The Capstone Project has been designed to achieve the following objectives:

1. Integration of Analytical Tools

- Learners consolidate Excel analysis, Python-based EDA, SQL querying, Power BI dashboards, inferential statistics, and simple linear regression within a single structured project.

2. Hypothesis-Driven Thinking

- Business questions are converted into testable hypotheses and validated using probability assessment and statistical tests.

3. Structured Data Handling & Documentation

- Learners maintain data dictionaries, preprocessing logs, version-controlled repositories, and structured analytical notes for reproducibility.

4. Insight Translation & Recommendation Development

- Analytical results are transformed into actionable business recommendations supported by quantitative evidence.

5. Professional Communication & Viva Readiness

- Learners prepare comprehensive reports, presentation decks, and confidently defend their analytical approach and outcomes during presentation days.

Project Workflow and Implementation Steps

The Capstone Project is divided into 11 structured phases followed by Presentation Days (1–4). Each phase builds progressively on the previous one, ensuring a structured and scaffolded

journey from dataset selection to statistical validation, regression analysis, business recommendations, and final showcase.

Each sprint has clearly defined learning objectives, deliverables, and video requirements. Learners are expected to complete each phase sequentially to ensure methodological consistency and analytical continuity throughout the project.

Phase 1: Kick-off & Project Allocation

Learners begin by understanding the capstone structure, expectations, sprint flow, and evaluation criteria. Teams are formed, roles are confirmed, and a dataset/topic is selected from the approved list.

Key Steps

- Review project goals, assessment criteria, and sprint flow.
- Select the dataset or problem from the given list.
- Document initial dataset details and intended direction.
- Complete readiness checks and clarify all requirements.

Phase 2: Data Familiarisation, Cleaning & Version Control

Learners explore the dataset's structure, perform structured preprocessing, and store the cleaned dataset in appropriate formats. Version control is established using Git for collaboration and documentation.

Key Steps

- Explore dataset fields, data types, and distributions.
- Identify and handle missing values, duplicates, and outliers.
- Create a data dictionary and preprocessing log.
- Store cleaned dataset in MySQL/SQLite/CSV/Excel formats.
- Initialize and maintain a structured Git repository with commits.

Phase 3: Business Problem Definition & Hypothesis Framing

Learners translate dataset understanding into structured business questions and testable hypotheses aligned with stakeholder context.

Key Steps

- Identify key problem areas within the dataset.
- Frame clear business and analytical questions.
- Convert questions into structured, testable hypotheses.
- Conduct preliminary data checks to validate feasibility.
- Document refined problem statement and hypothesis plan.

Phase 4: Excel Analysis & Dashboards

Learners apply advanced Excel functions and analytical tools to summarise data and design a storytelling-focused dashboard.

Key Steps

- Use Excel functions such as XLOOKUP and conditional formulas.
- Build pivot tables with grouping, filtering, and summarisation.
- Create pivot charts including pie, line, bar, scatter, histogram, and boxplots.
- Compute descriptive statistics such as central tendency, distribution, skewness, and correlation.
- Design an Excel dashboard combining visuals and insights into a coherent narrative.

Phase 5: Python-Based Exploratory Data Analysis

Learners perform exploratory data analysis using Python to deepen analytical understanding and refine project direction.

Key Steps

- Generate summary statistics and explore distributions.
- Analyse correlations, trends, and group-level behaviours.
- Create histograms, scatterplots, boxplots, and heatmaps.
- Highlight patterns influencing hypothesis outcomes.
- Document insights and refine analytical direction accordingly.

Phase 6: SQL-Based Analysis for Insight Generation

Learners apply SQL queries to extract structured insights and create summary tables for analytical interpretation.

Key Steps

- Write SQL queries using GROUP BY, WHERE, HAVING, SUM, AVG, and COUNT.
- Perform joins and filtering to answer analytical questions.
- Generate derived tables or Common Table Expressions (CTEs).
- Document SQL outputs and corresponding analytical insights.

Phase 7: Data Visualization and Dashboard using Power BI

Learners develop interactive dashboards using Power BI and define KPI metrics aligned with project objectives.

Key Steps

- Export and prepare data for Power BI.
- Define key performance indicators (KPIs).

- Apply DAX formulas for analytical calculations.
- Create interactive dashboards with filters and visuals.
- Narrate insights using structured storytelling.

Phase 8: Statistical Validation of Assumptions

Learners apply inferential statistical methods to validate predefined hypotheses using data-driven evidence.

Key Steps

- Perform probability assessments where required.
- Conduct hypothesis testing using z-test and t-test.
- Document test statistics and p-values.
- Interpret results to reject or fail to reject hypotheses.
- Summarise validation outcomes clearly.

Phase 9: Simple Linear Regression Analysis, Evaluation & Interpretation

Learners prepare the dataset for regression analysis, build a simple linear regression model, and evaluate its performance using appropriate metrics.

Key Steps

- Perform feature engineering and required transformations.
- Standardise/scale variables where necessary.
- Split dataset into train–validation–test sets.
- Build a simple linear regression model and generate predictions.
- Evaluate model performance using relevant metrics.
- Document modelling steps, results, and justification.

Phase 10: Business Insights, Recommendations & Future Scope

Learners translate analytical findings into structured business insights and actionable recommendations.

Key Steps

- Consolidate insights from Excel, Python, SQL, Power BI, statistical tests, and regression outputs.
- Develop clear, data-driven recommendations.
- Highlight project limitations and challenges encountered.
- Define future scope and next steps for extended analysis.

Phase 11: Documentation & Presentation Preparation

Learners compile all work into structured documentation and prepare for final showcase and viva sessions.

Key Steps

- Develop and finalise comprehensive written report.
- Draft professional presentation slide deck.
- Ensure narrative alignment between insights and visuals.
- Practise delivery and prepare for viva-style questions.

Final Presentation (Showcase & Viva Voce)

- Learners formally present their project findings to the review committee across scheduled presentation days.
- Deliver the final PPT presentation and demonstrate dashboards and outputs.
- Defend methodology, statistical validation, and regression approach during viva sessions.
- Engage in peer learning and respond to mentor questions.
- Receive structured evaluation and complete the capstone cycle.

Integrate GenAI for Storytelling (Optional but Recommended)

- Learners may responsibly use tools such as ChatGPT or similar platforms to:
 - Refine business problem statements and hypothesis articulation.
 - Generate structured summaries for documentation.
 - Draft presentation scripts aligned with analytical findings.
- Review outputs critically to ensure accuracy and alignment with project work.

Submit Final Deliverables

- Your final submission should include:
 - Cleaned dataset (CSV / SQL / Excel format).
 - Python notebook (.ipynb or .py), SQL scripts, Excel workbook, Power BI file.
 - Statistical validation documentation and regression outputs.
 - Final written report.
 - Presentation slide deck for showcase and viva.

Capstone Project Checklist

A structured checklist to help learners track progress across all sprints. Each item should be completed before moving to the next phase to ensure continuity and methodological discipline.

Phase 1: Kick-off & Project Allocation

- ☐ Dataset/topic selected from approved list and documented
- ☐ Team roles confirmed and responsibilities defined
- ☐ Project goals, sprint roadmap, and evaluation rubric reviewed

- ☐ Dataset downloaded and initial details captured

Phase 2: Data Familiarisation, Cleaning & Version Control

- ☐ Dataset structure explored (fields, data types, distributions)
- ☐ Missing values, duplicates, and outliers handled systematically
- ☐ Data dictionary and preprocessing log created
- ☐ Cleaned dataset stored in SQL / CSV / Excel format
- ☐ Git repository initialised with structured commits

Phase 3: Business Problem Definition & Hypothesis Framing

- ☐ Business problem statement drafted and refined
- ☐ Analytical questions framed clearly
- ☐ Testable hypotheses defined and documented
- ☐ Preliminary feasibility checks completed

Phase 4: Excel Analysis & Dashboards

- ☐ Excel functions (e.g., XLOOKUP, conditional formulas) applied.
- ☐ Pivot tables created with grouping and filtering.
- ☐ Pivot charts (pie, line, bar, scatter, histogram, boxplot) generated.
- ☐ Descriptive statistics computed (central tendency, distribution, skewness, correlation).
- ☐ Excel dashboard designed with storytelling elements.

Phase 5: Python-Based Exploratory Data Analysis

- ☐ Summary statistics generated in Python
- ☐ Distributions, correlations, and group behaviours analysed
- ☐ Visualisations created (histograms, scatterplots, boxplots, heatmaps)
- ☐ EDA insights documented and linked to hypotheses

Phase 6: SQL-Based Insight Generation

- ☐ SQL queries written using joins, filters, and aggregations
- ☐ GROUP BY, WHERE, HAVING, SUM, AVG, COUNT applied appropriately
- ☐ Derived or summary tables created
- ☐ SQL insights documented with supporting outputs

Phase 7: Power BI Dashboard & KPI Narration

- ☐ KPI metrics defined and documented
- ☐ DAX formulas applied for analysis
- ☐ Interactive dashboard created with filters and visuals
- ☐ Visual insights summarised with storytelling

Phase 8: Statistical Validation of Assumptions

- ☐ Probability assessments performed where required
- ☐ Hypothesis testing conducted (z-test, t-test)
- ☐ Test statistics and p-values documented
- ☐ Results interpreted (reject / fail to reject hypotheses)
- ☐ Statistical validation summary created

Phase 9: Simple Linear Regression Analysis & Evaluation

- ☐ Feature engineering and required transformations completed
- ☐ Variables standardised/scaled where required
- ☐ Dataset split into train-validation-test sets
- ☐ Simple linear regression model built and predictions generated
- ☐ Evaluation metrics calculated and interpreted
- ☐ Modelling steps and justification documented

Phase 10: Business Insights, Recommendations & Future Scope

- ☐ Key insights consolidated from Excel, Python, SQL, Power BI, statistical tests, and regression outputs
- ☐ Data-driven recommendations drafted
- ☐ Limitations and challenges documented
- ☐ Future scope and next steps outlined

Phase 11: Documentation & Presentation Preparation

- ☐ Final written report compiled
- ☐ Presentation deck drafted and structured
- ☐ Narrative aligned between documentation and visuals
- ☐ Presentation rehearsed for clarity and timing

Final Presentation (Showcase & Viva Voce)

- ☐ Final PPT delivered to evaluators
- ☐ Methodology, statistical validation, and regression approach defended during viva
- ☐ Viva responses provided confidently
- ☐ Feedback received and documented
- ☐ Project journey reflected upon and formally concluded

Deliverables and Timelines

The Capstone Project is structured across 11 phases followed by Presentation Days (1–4). Each deliverable represents a measurable milestone aligned with sprint objectives. These

milestones demonstrate your ability to clean data, perform structured analysis using Excel, Python, SQL, and Power BI, apply inferential statistics, conduct simple linear regression analysis, and translate findings into business recommendations.

The table below outlines the sprint-wise deliverables and video requirements.

Deliverable	Details	Sprint	Video (Y/N) & Brief
1. Topic Selection & Project Acknowledgement	Selected dataset/topic from approved list, evaluation rubric acknowledgement, initial dataset review and documentation	Sprint 1	No video required
2. Dataset Understanding Summary & Cleaned Dataset	Dataset structure summary, data dictionary, preprocessing log, cleaned dataset (SQL/CSV/Excel), Git repository with structured commits	Sprint 2	Yes — 3–4 min explainer: dataset structure, issues identified, cleaning logic
3. Business Problem Statement & Hypotheses Document	Business problem statement, analytical questions, structured and testable hypotheses	Sprint 3	Yes — 3–4 min explaining problem framing and hypothesis logic
4. Excel Analysis Workbook & Dashboard	Excel workbook with advanced functions, pivot tables, descriptive statistics, charts, storytelling dashboard	Sprint 4	Yes — 3–4 min walkthrough explaining dashboard insights
5. Python EDA Notebook	Summary statistics, visualisations (histograms, scatterplots, boxplots, heatmaps), EDA insights note	Sprint 5	Yes — 3–4 min summarising Python EDA findings
6. SQL Query Scripts & Structured Insights	SQL scripts using joins, filters, aggregations; derived/summary tables; documented insights	Sprint 6	Yes — 3–4 min explaining at least 2 SQL queries and outcomes
7. Power BI Dashboard & KPI Sheet	Interactive Power BI dashboard, KPI definition sheet, DAX calculations, visual insights summary	Sprint 7	Yes — 4–5 min dashboard demo showing filters, KPIs, and insights
8. Statistical Validation Report	Probability assessments, z-test and t-test outputs, test statistics, p-values, hypothesis validation summary	Sprint 8	Yes — 3–4 min explaining 2 key statistical tests and interpretations
9. Simple Linear Regression Output & Evaluation	Feature-engineered dataset, regression model, predictions, evaluation metrics, modelling justification	Sprint 9	Yes — 4–5 min explaining model choice, features, and evaluation metrics
10. Business Insights & Strategic	Consolidated analytical findings, actionable	Sprint 10	Yes — 3 min executive-style recommendations presentation

Recommendations	recommendations, limitations, future scope		
11. Final Report & Presentation Deck	Comprehensive written report, structured documentation, draft PPT for showcase	Sprint 11	Yes — Recorded capstone showcase presentation
12. Capstone Showcase & Viva (Presentation Days 1–4)	Final PPT presentation, viva responses, methodology defence, peer learning participation	Presentation Days	No additional video (live showcase & viva)

Guidelines for Success

The capstone journey is designed to help learners apply structured analytical methods in a disciplined and professional manner. Following these guidelines will ensure systematic execution, well-documented outputs, and high-quality deliverables across all sprint phases.

1. Planning & Preparation

- Define a clear and focused business problem aligned with the selected dataset.
- Review sprint objectives and deliverables before beginning each phase.
- Follow the structured sequence of phases to maintain analytical continuity.
- Plan workload in alignment with sprint timelines and video requirements.

2. Data Handling & Quality

- Explore dataset structure thoroughly before beginning analysis.
- Handle missing values, duplicates, and outliers systematically and document decisions.
- Maintain a detailed data dictionary and preprocessing log.
- Store cleaned datasets in approved formats (SQL / CSV / Excel) and maintain version control using Git.

3. Analytical Rigor

- Apply descriptive statistics in Excel and Python to understand distributions and relationships.
- Use SQL queries strategically to generate structured summary tables and insights.
- Conduct inferential statistical tests (z-test, t-test, probability assessments) to validate hypotheses.
- Perform simple linear regression with proper feature preparation and interpret outputs carefully.
- Use appropriate evaluation metrics relevant to regression analysis and document justification.

4. Tool & Technology Use

- Use Excel functions, pivot tables, and charts to generate structured dashboards.
- Leverage Python for exploratory data analysis and visual summaries.
- Apply SQL operations such as joins, filters, and aggregations for insight extraction.
- Build interactive dashboards in Power BI using defined KPIs and DAX calculations.
- Use GenAI tools responsibly for documentation support while validating all outputs independently.

5. Collaboration & Documentation

- Maintain structured documentation for each sprint phase.
- Ensure Git repositories are updated with meaningful and organised commits.
- Clearly document modelling steps, statistical outputs, and analytical interpretations.
- Align all insights consistently across notebooks, dashboards, and reports.

6. Communication & Presentation

- Translate analytical findings into clear, business-focused insights.
- Ensure dashboards and reports highlight interpretation, not just numerical outputs.
- Keep sprint videos concise and focused on key findings and logic applied.
- Rehearse presentation delivery to ensure clarity, confidence, and structured narration.

7. Professionalism & Integrity

- Adhere strictly to ethical data handling and academic integrity standards.
- Cite dataset sources appropriately in documentation and presentation materials.
- Ensure originality in analysis, coding, and documentation.
- Approach each sprint as a structured professional assignment with attention to quality and completeness.

Success in this capstone project depends on maintaining analytical discipline, structured documentation, tool proficiency, and clear communication. Each sprint builds toward a final showcase that reflects both technical competence and professional readiness.

Evaluation Criteria

Given below are the criteria against which your project will be assessed. You are encouraged to conduct a self-assessment. Each parameter may be rated on a scale of 1–5 as per the rating scale:

Category	Criteria
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1. Planning & Project Alignment	1.1 Selected an appropriate dataset and clearly aligned it with a defined business problem.
	1.2 Demonstrated clarity in project objectives and adherence to sprint structure.
	1.3 Followed the defined workflow sequentially across all phases.
2. Data Selection, Understanding & Preprocessing	2.1 Explored dataset structure (fields, data types, distributions) thoroughly.
	2.2 Created a data dictionary and handled missing values, duplicates, and outliers systematically.
	2.3 Maintained preprocessing logs and stored cleaned data in approved formats (SQL/CSV/Excel).
3. Exploratory Data Analysis (Excel & Python)	3.1 Applied descriptive statistics and summary measures to explore key variables.
	3.2 Identified patterns, correlations, and trends using visual and statistical summaries.
	3.3 Documented exploratory insights clearly and linked them to hypotheses.
4. Data Visualization & Dashboard Storytelling	4.1 Designed structured dashboards using Excel and Power BI.
	4.2 Defined KPIs and aligned visuals with the project narrative.
	4.3 Translated analytical findings into clear and engaging insights.
5. SQL-Based Insight Generation	5.1 Applied joins, filters, grouping, and aggregations appropriately.
	5.2 Generated structured summary tables to support analysis.
	5.3 Documented SQL outputs and interpretations clearly.
6. Statistical Validation	6.1 Applied probability assessments and conducted z-test and t-test correctly.
	6.2 Reported test statistics and p-values accurately.
	6.3 Interpreted statistical outcomes logically in relation to predefined hypotheses.
7. Simple Linear Regression Analysis & Evaluation	7.1 Performed feature preparation and necessary transformations appropriately.
	7.2 Built and evaluated a simple linear regression model using suitable metrics (e.g., R^2 , error-based measures).
	7.3 Interpreted regression results in a business and analytical context.
8. Project Documentation & Presentation	8.1 Prepared a structured and professional final report with clear methodology and findings.

	8.2 Designed a coherent presentation aligned with insights and recommendations.
	8.3 Delivered a clear and confident presentation during showcase and viva.

Project Topics

Choose from the following real-world themes for your capstone project. Each topic is designed to help you apply descriptive statistics, hypothesis testing, SQL-based analysis, dashboard storytelling, and simple linear regression on realistic datasets.

Note: These synthetic datasets are structured to mirror real-world analytical scenarios, enabling hands-on practice in exploratory analysis, statistical validation, regression modelling, and business intelligence reporting.

Topic 1: Analyse E-Commerce Sales to Improve Customer Conversion & Retention (RetailTech)

Topic 2: Assess Fitness App Usage & Calorie Burn Patterns to Drive Personalized Engagement (HealthTech)

Topic 3: Evaluate Medicine Supply Chain Demand & Inventory Performance for Better Planning (MedTech)

Topic 4: Analyse Social Media Engagement to Uncover Brand Performance Insights (SocialTech)

Topic 5: Analyse OTT Platform Viewing Patterns to Support Data-Driven Content Decisions (MediaTech)

Topic 6: Explore Gamer Behavior & Monetization Patterns to Optimize In-Game Engagement (GameTech)

Topic 7: Analyse Food Delivery Demand Across Regions to Inform Market Expansion & Logistics (FoodTech)

Topic 8: Evaluate Crop Production & Yield Data Across India to Strengthen Agricultural Planning (AgriTech)

Topic 9: Analyse Campus Placement & Hiring Data to Understand Talent Pipeline & Predict Success Traits (EduTech)

Topic 10: Analyse Operational Data to Improve Logistics & Warehousing Efficiency (LogiTech)

Conclusion

The Capstone Project represents the culmination of your applied analytics learning journey. It brings together the structured use of Excel, Python, SQL, Power BI, inferential statistics, and simple linear regression within a clearly defined sprint-based framework.

By following the structured multi-phase workflow and completing each deliverable — including documentation, dashboards, statistical validation, regression outputs, and recorded presentations — you demonstrate both analytical competence and professional readiness.

This project is not limited to technical execution. It requires disciplined thinking — defining clear business questions, validating assumptions using statistical methods, interpreting regression outputs carefully, and communicating findings with clarity and impact.

As you progress toward the final showcase, remember that:

Every sprint is a structured milestone — complete it with accuracy and attention to detail.

Documentation, interpretation, and clarity are as important as calculations and visuals.

The final presentation reflects both analytical depth and professional confidence.

Key Takeaway:

This capstone is your opportunity to convert structured learning into demonstrated capability. Approach each phase with discipline and analytical focus, and it will position you as a confident data-driven problem solver capable of delivering structured, evidence-based insights.

✦ **Embrace the journey you've completed, recognize the skills you've mastered, and step forward with confidence as a data-driven problem solver and storyteller.**